



Puduvai **Innovation** Competition 2026

TITLE: AI-Assisted Smart Bus Decision System for Reducing Commuter Uncertainty

ID: IC2026-251

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Problem Statement:

- Daily bus users often face confusion due to lack of clear and real-time information. They do not know which bus will arrive, when it will come, or whether it will take them to their destination.
- This becomes even more difficult for beginners and students who are not familiar with routes.
- Additionally, there is no reliable way to know the crowd level inside the bus, which leads to uncomfortable and inefficient travel decisions.

Proposed Solution:

- **Autonomous IoT Kit:** A "fit-and-forget" hardware unit using ESP32 and IR Break-Beam sensors to count passengers automatically at both doors.
- **Intelligent Edge Logic:** Passenger counting only activates when $\text{Speed} = 0$, preventing false counts while the bus is moving.
- **Passive Powering:** Draws power directly from the bus battery via a Buck Converter—no charging or human intervention needed.
- **Decision App:** Features a "Regular Bus Pin" that notifies users: "Your bus is 10 mins away; it's 90% full. Suggestion: Wait 6 mins for the next empty bus."

Target Market & Value Proposition:

- **Target Market:**

- Primary Users: 50,000+ college students and daily commuters in the UT.
- The Customers (Who pays): The Government of Puducherry (PRTC / Smart City Mission) and Private Bus Owner Associations.

- **Value Proposition (Why use it):**

- For Students: Eliminates "Wait-Anxiety" and ensures a safer, seated journey.
- For Government: Provides a Digital Audit of free student-pass usage and saves fuel by optimizing routes based on real demand.
- For Owners: Stops "Revenue Leakage" by providing an independent daily passenger count.

Revenue Model & Scalability:

Our project follows a SaaS (Software-as-a-Service) Freemium model, where basic features are free and advanced services are paid.

Free Version:

Basic bus tracking and limited crowd information for public users.

Pro Version (₹1,000–₹3,000/month):

For private operators and colleges with real-time tracking, alerts, and reports.

Enterprise Version (₹2–5 Lakhs/year):

For government and smart city projects with AI predictions, analytics, and API integration.

Who Pays:

Government transport departments, private bus operators, colleges, and smart city authorities.

Revenue Sources:

Subscriptions, advertisements, data insights, and API access.

Go-To-Market:

Pilot , direct demos, and partnerships with transport agencies.

Execution Readiness:

Technical Readiness: Hardware logic and component list (ESP32/IR/GPS) finalized; estimated cost is ₹1,800 per unit.

Market Entry Plan:

Pilot: Install kits on one major route (e.g., Pondy Bus Stand to University) for 14 days.

Data Proof: Present the "Route Efficiency & Passenger Audit" report to the Transport Commissioner.

Resources Needed: ₹20,000 for a 10-bus pilot and a simple Flutter-based app prototype.